Practical 3

Name: Devarsh Dalwadi
Roll No.: 23BCE051

Objective:

Implement a Multi-Container Application Using Docker Compose. The application should include networking to communicate using a custom bridge network.

Implementation Details:

We have implemented a multi-container application using Python Flask and Redis. The Flask web app increments a hit counter stored in the Redis cache. Both containers communicate over a custom bridge network called "app-network".

Source Code:

File: docker-compose.yml

```
version: "3.9"
services:
  web:
    build: .
    ports:
      - "5000:5000"
    networks:
      - app-network
    depends_on:
      - redis
  redis:
    image: "redis:alpine"
    networks:
      - app-network

networks:
  app-network:
    driver: bridge
```

File: app.py

```
import time
import redis
from flask import Flask

app = Flask(__name__)
cache = redis.Redis(host='redis', port=6379)

def get_hit_count():
    retries = 5
```

```
    while True:
        try:
            return cache.incr('hits')
        except redis.exceptions.ConnectionError as exc:
            if retries == 0:
                raise exc
            retries -= 1
            time.sleep(0.5)

@app.route('/')
def hello():
    count = get_hit_count()
    return f'Hello from Docker! I have been seen {count} times.\n'

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

File: Dockerfile

```
FROM python:3.9-alpine
WORKDIR /code
ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
COPY requirements.txt requirements.txt
RUN pip install -r requirements.txt
EXPOSE 5000
COPY . .
CMD ["flask", "run"]
```

File: requirements.txt

```
flask
redis
```

Execution & Screenshots:

1. Starting the application



2. Application Output (Browser/Curl)

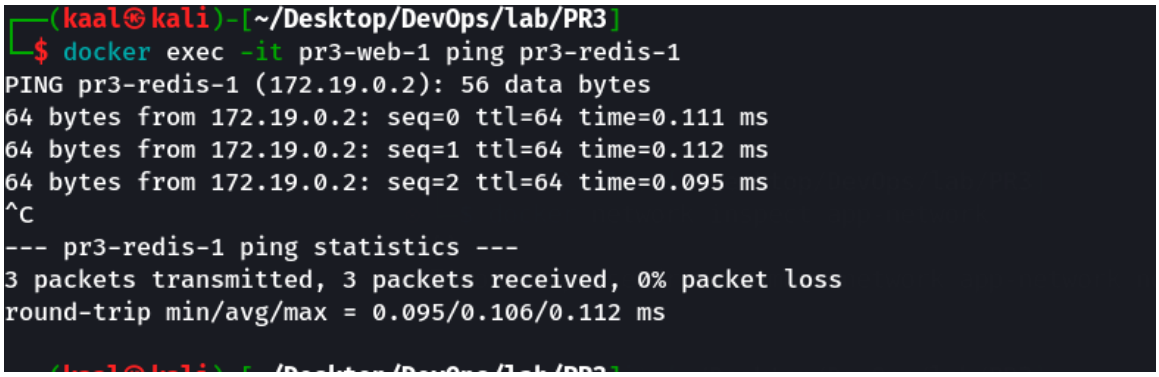


3. Inspecting the Bridge Network

Command run: `docker network inspect pr3_app-network`

```
┌──(kaal㉿kali)-[~/Desktop/DevOps/lab/PR3]
└─$ docker network inspect pr3_app-network
[
    {
        "Name": "pr3_app-network",
        "Id": "18b8b8f9ef9c6b6136321ee32e97c9a8683dbda97be3f1e3fe312c2898712a6e",
        "Created": "2026-09-01T15:14:10.060248435+05:30",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "172.19.0.0/16",
                    "Gateway": "172.19.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
```

Communication between containers.


┌──(kaal㉿kali)-[~/Desktop/DevOps/lab/PR3]
└─$ docker exec -it pr3-web-1 ping pr3-redis-1
PING pr3-redis-1 (172.19.0.2): 56 data bytes
64 bytes from 172.19.0.2: seq=0 ttl=64 time=0.111 ms
64 bytes from 172.19.0.2: seq=1 ttl=64 time=0.112 ms
64 bytes from 172.19.0.2: seq=2 ttl=64 time=0.095 ms
^C
--- pr3-redis-1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.095/0.106/0.112 ms